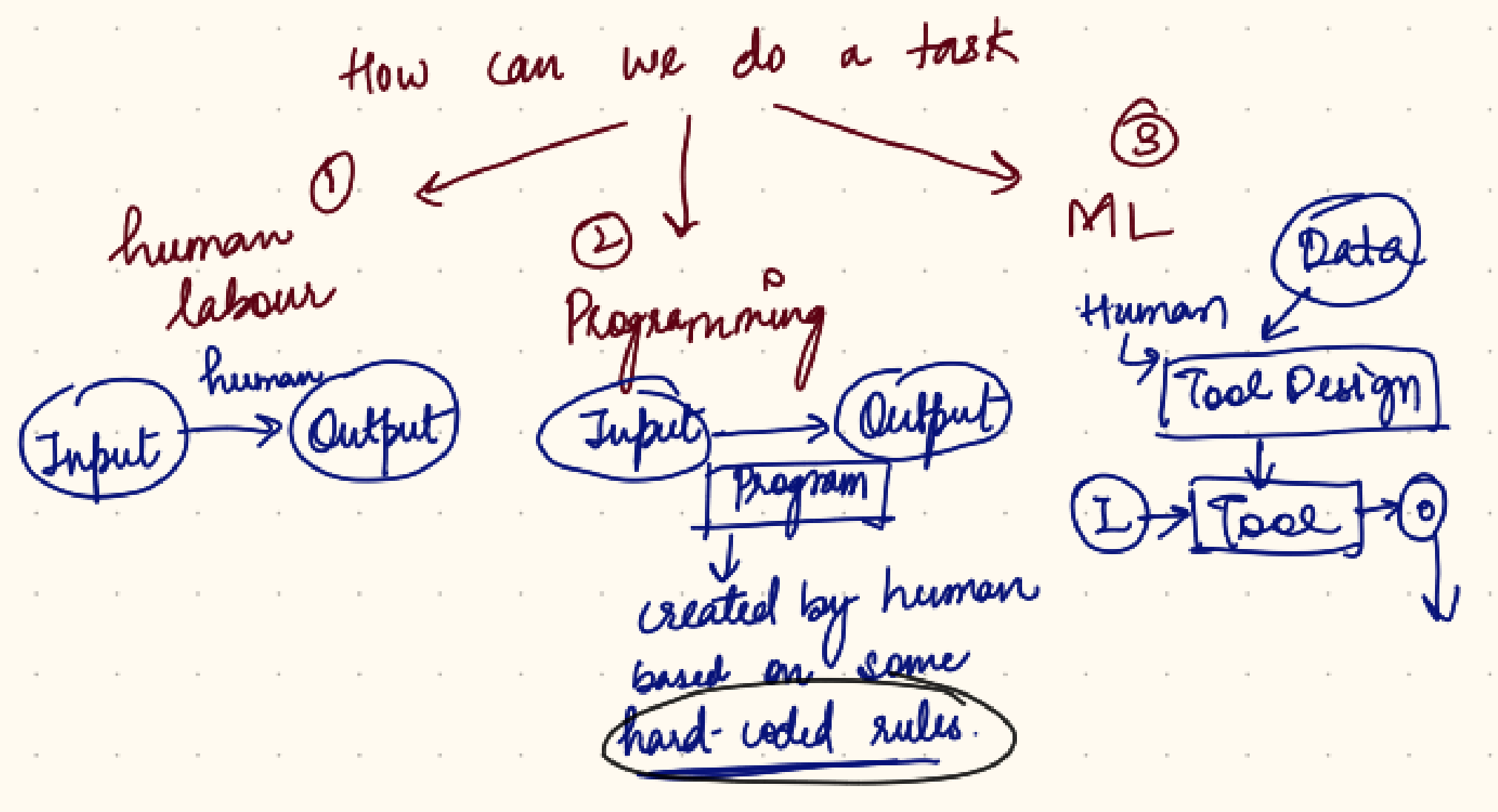
# Machine Learning Foundations

## Week-1 , Session-1

1. What is ML?
   - → study of algorithms
   - → improves automatically through data.

How can we do a task



₹ x  y% → formula → output

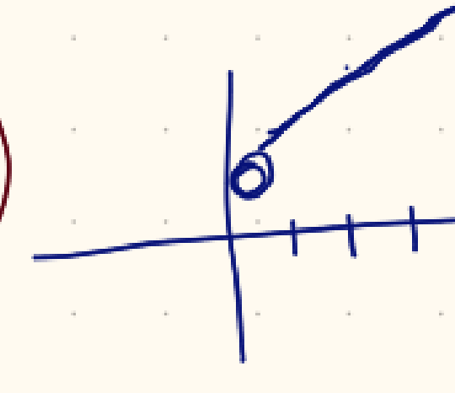
To solve a problem using ML →

1) Data
2) some structural idea on the rules

| Rule | Formula | fits | Price @ 5BHK | Price@ 1BHK | |
|---|---|---|---|---|---|
| A: straight line | 2x rooms | ✓ | 10 | 2 | |
| B: line + wiggle | 2xrooms + (room-2)x (room-3)x (room-4) | ✓ | 16 | -4 → | X |
| C: lookup Table | | | 0 | 0 → | X |

~~we~~ know # rooms

| rooms | price |
|---|---|
| 2 | 4 |
| 3 | 6 |
| 4 | 8 |
| other | 0 |



Data → $\begin{bmatrix} ② \text{ rooms} \to 4 \text{ cr} \\ 3 \text{ rooms} \to 6 \text{ cr} \\ 4 \text{ rooms} \to 8 \text{ cr} \\ 6 \text{ rooms} \to \text{10 cr} \end{bmatrix}$ 0

x

4 = 0
6 = 0
8 = 0
12 = ②

Fit → loss on training data

$$\boxed{\text{cost} = a \times \underline{\text{rooms}} + b \times \text{dist from school} - c \times \text{dist from traffic prone road}}$$

$$\underline{\text{cost}} = 3\underline{a} + \underline{b} - 5\underline{c}$$

| rooms | dist$_1$ | dist$_2$ |
|---|---|---|
| — | — | — |
| — | — | — |

model → $\boxed{a, b, c}$ using → predict cost

$$\text{cost} = \underline{a}\,\text{room}^2 + \underline{b}\,\text{room}$$

| Data | Model | Learning Algo |
|---|---|---|
| collection of vectors X | simplification approximations $y = 3x_1 + 5x_2 - 2x_3$ | trying to fit (How) a chosen family of fun^n using the data ↓ linear |

(How) → minimizing Loss

features $\begin{bmatrix} \text{rooms} \\ d\ \text{school} \\ d\ \text{traffic} \end{bmatrix} \to \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}$

target → price → 7 (y)

| Chennai | Mumbai | Delhi |
|---|---|---|
| (1, | 0, | 0) |
| (0, | 1, | 0) |
| (0, | 0, | 1) |

$$\boxed{y = \underline{a}\,x_1 + \underline{b}\,x_2 + \underline{c}\,x_3}$$

$x_1, x_2, x_3$ → features

trying to learn/fit (a, b, c) using data

parameters ← (a, b, c)

(10, 2, 3) →

(5, -1, 2) →

Diff a,b,c give diff models.
Learning algo searches over (a,b,c) to find the best model.
↳ based on some loss function.

Models
- → Predictive
  - → regression (predict a number)
  - → classification (predict a label)
- → Probabilistic → how likely is something? (probability)

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## Learning

### Supervised

Data + labels

[ ] + ___

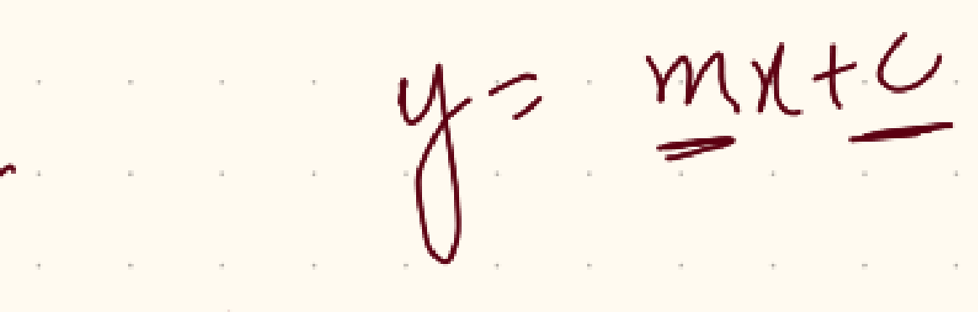
X + (y)

↳ curve-fitting

$X ; y \to \{(x^i, y^i)\}$

Aim → $\boxed{f(x_i) \approx y}$

Regression → y is a number
Classifica^n → y is a label



$y = \underline{m}x + \underline{c}$

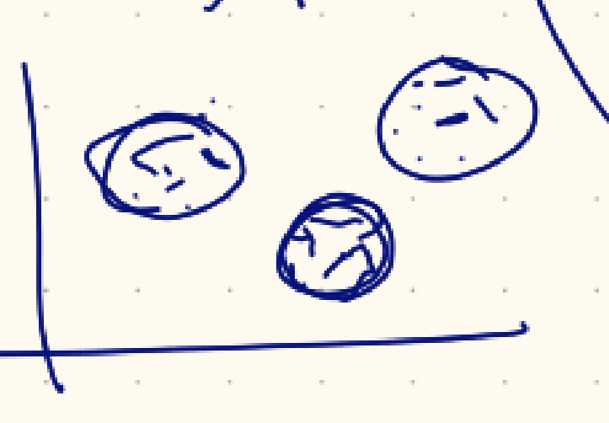
### Unsupervised

Data (no labels)

[ : ]

X

$x_1$
$x_2$
⋮
$x_n$



Loss → $\min \frac{1}{n} \sum_i (f(x^i) - y^i)^2$ SSE

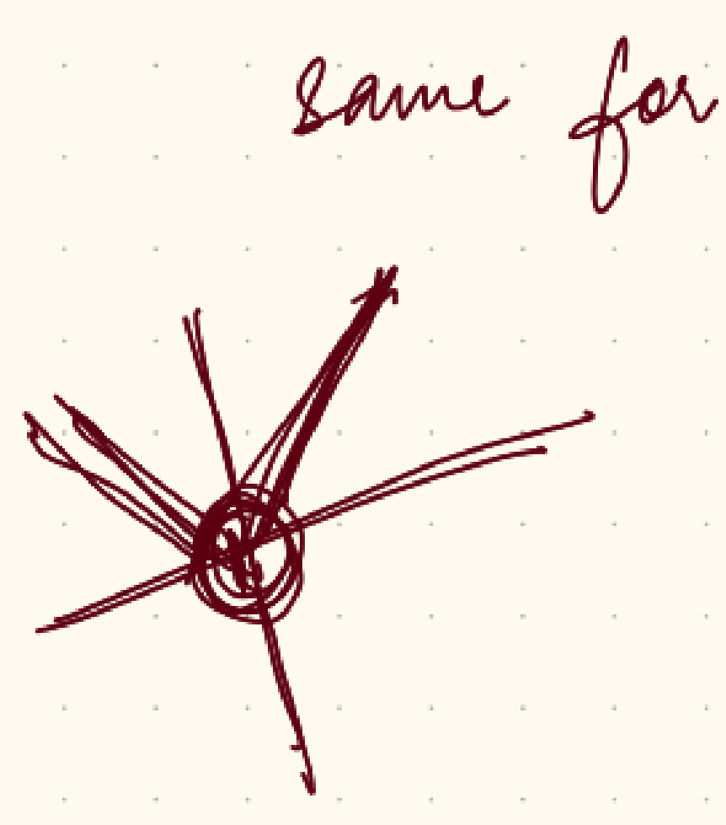
$f(x_1) \approx m x_1 + c$ for $(x_1, y_1)$

$\boxed{f(x_1) - y_1}$ → 0.1 0.01
→ 5 25

same for $x_2$ — — $x_n$

5

0.1



$$\boxed{y = w^T x + b}$$